

CHE 205 - Computational Methods in Chemical Engineering - Fall 2019

Required Text: Numerical Methods for Chemical Engineers Using Excel, VBA, and MATLAB, by V.J. Law. ISBN: 978-1-4665-7534-9, CRC Press, 2013.

Topics to be Covered:

- Ch. 1: Roots of a single nonlinear equation
- Ch. 2: Visual Basic for applications programming
- Ch. 3: Linear algebra and systems of linear equations
- Ch. 4: Numerical differentiation and integration
- Ch. 5: Ordinary differential equations (initial value problems)
- Ch. 6: Ordinary differential equations (boundary value problems)
- Ch. 8: Partial differential equations
- Ch. 7: Regression analysis and parameter estimation
- Ch. 9: Linear programming, nonlinear programming, nonlinear equations, and nonlinear regression using Solver
- Ch. 10.: Introduction to Matlab, basics, programming language statements, function arguments, and plotting.

Supplemental Materials: If you need to learn more about VBA programming and Matlab, you may find many tutorials on the internet. You can also consult with the following websites and books:

- http://excelvbatutor.com/vba_tutorial.html
- http://www.homeandlearn.org/basic_mathematical_operators.html
- <http://www.automateexcel.com/vba-code-examples>
- Computational Methods for Engineers with Matlab Applications, by James B. Riggs, ISBN 0-9669601-5-7, Ferret Publishing, Austin, TX, 2013
- Numerical Computing with Matlab, by Cleve B. Moler, ISBN ISBN-10: 0898715601, Society for Industrial and Applied Mathematics (June 2004)
- Matlab Machine Learning, by Michael Paluszec and Stephanie Thomas, ISBN-13: 978-1-4842-2250-8, Apress; 1st ed. edition (December 29, 2016)

- Quick and free collection of MATLAB tutorials, videos, training, and documentation (<https://www.mathworks.com/support/learn-with-matlab-tutorials.html>)

Objectives: By taking this course, you will be able to

1. write VBA and Matlab scripts for most algorithms in chemical engineering.
2. solve many mathematical problems in thermodynamics and transport numerically using spreadsheets and Matlab.

Prerequisite(s): Credit or concurrent registration in CHE 201, and MATH 210